

Exosomes based strategies for brain drug delivery

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Abstract

Exosome application has emerged as a promising nanotechnology discipline for various diseases therapeutics and diagnoses. Owing to the natural properties of efficient drug delivery, higher biocompatibility, facile traversing of physiological barriers, and subtle side effects, exosomes shorten their way to clinical translation. Exosomes are nanoscale membrane-bound vesicles primarily involved in intercellular communication and exhibit natural blood-brain barrier (BBB) traversing ability, which enables their application as drug delivery vehicles for brain diseases treatment. Herein, we highlight recent exosome-based drug delivery endeavors for neurodegenerative diseases and brain cancer therapy, summarize the obstacles and future directions in clinical translation.